LECTURE SERIES · DATABASE SYSTEMS

# Physical Storage Systems

*Where the Data Actually Lives — Disks, Blocks & RAID*

**Md. Rakib Trofder**

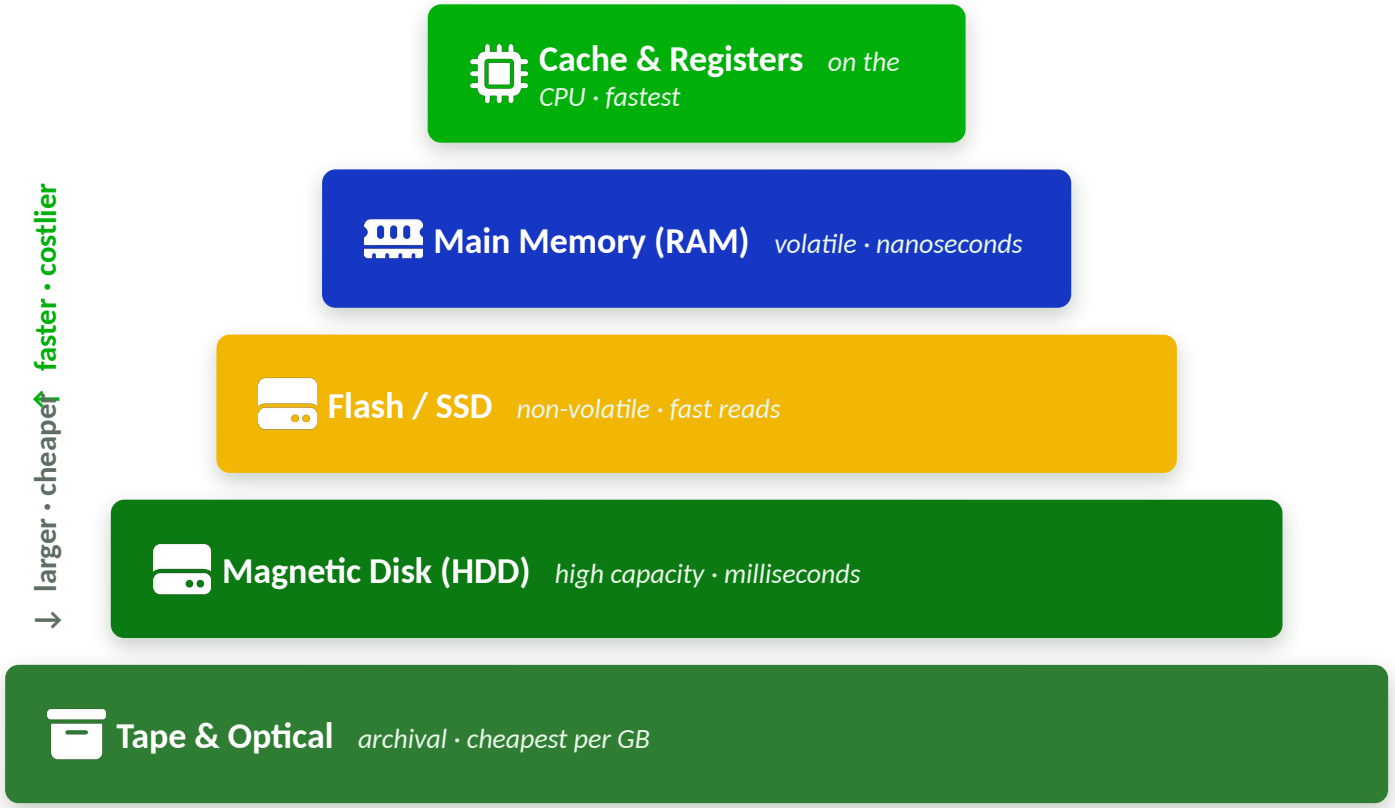
Lecturer, Bangladesh Army University of Science & Technology

Storage Hierarchy · Storage Structures · Striping · Mirroring · Parity · RAID Levels

PHYSICAL STORAGE SYSTEMS

# The Storage Hierarchy

Storage forms a hierarchy: the fastest media are the smallest and most expensive; capacity grows as speed falls.




### PRIMARY STORAGE

Volatile — contents vanish on power-off. The CPU works directly from here.

### SECONDARY STORAGE

Non-volatile — where the database files permanently live (SSD / HDD).

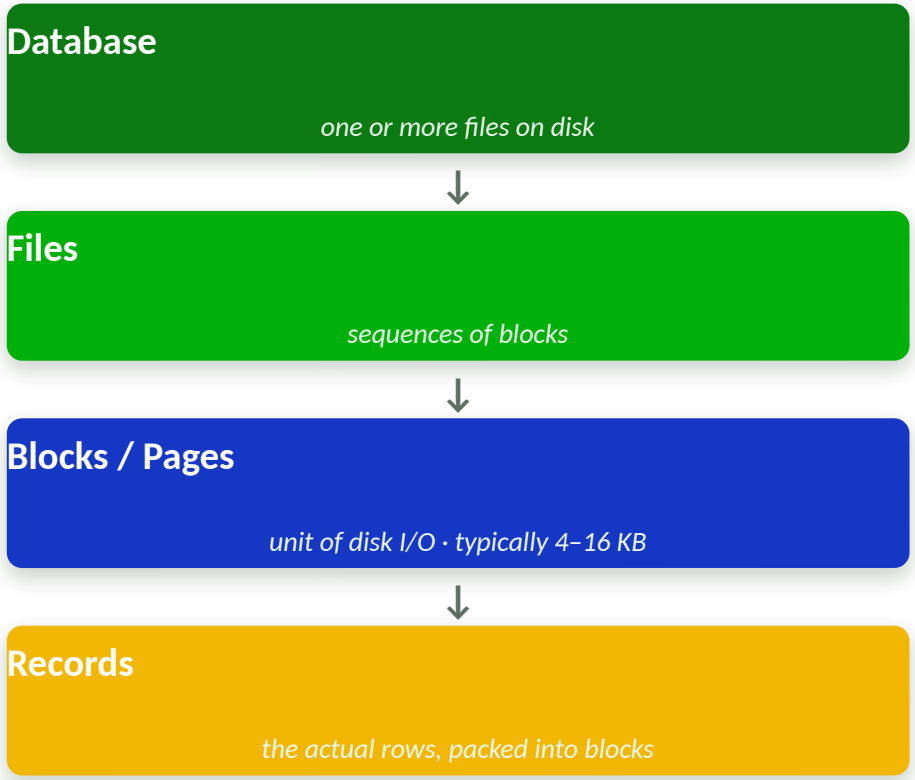
### TERTIARY STORAGE

Offline archives & backups: tape libraries, optical disks.

HOW FILES HOLD RECORDS

# Data Storage Structures

On disk, a database is a set of files, each divided into fixed-size blocks — and blocks are where records actually sit.




## FILE ORGANIZATIONS

### Heap

Records go wherever there is space — fast inserts, full scans to search.

### Sequential

Records kept sorted on a search key — great for ranges & ordered scans.

### Hashing

A hash of the key picks the block — near-instant equality lookups.

**Indexes** sit on top of these files to speed up lookups.

REDUNDANT ARRAY OF INDEPENDENT DISKS

# RAID — Arrays of Disks

**RAID** combines several physical disks into one logical unit — trading among three goals: **speed, capacity, and reliability.** It is built from three techniques:

## Striping

Split data across disks so they read & write in parallel.

`goal: SPEED`

## Mirroring

Keep a full duplicate of every disk.

`goal: RELIABILITY`

## Parity

Store checksums that can rebuild a lost disk.

`goal: RECOVERY, cheaply`

**RAID is not a backup:** it survives disk failure — not deletion, corruption, or disasters.

RAID 0 — PURE SPEED

# Data Striping

Striping chops data into blocks dealt across disks in turn. Reads and writes hit all disks in parallel — bandwidth multiplies.

**+ Full capacity**

every byte of every disk is usable

**+ Fastest reads & writes**

parallel I/O across all disks

**- Zero fault tolerance**

one disk dies → the whole array is lost

**Use for:** scratch space, caches, video editing.

**RAID 0 — file A striped over two disks**




*blocks alternate: A1→Disk1, A2→Disk2, A3→Disk1 …*

**capacity = 100% • tolerates 0 failures**

**Granularity:** stripe by bit, byte, or (usually) block.

RAID 1 — PURE RELIABILITY

# Mirroring

Mirroring writes every block to two (or more) disks at once. Lose a disk — the twin carries on without missing a beat.

**+ Survives disk failure**

service continues from the mirror

**+ Fast reads**

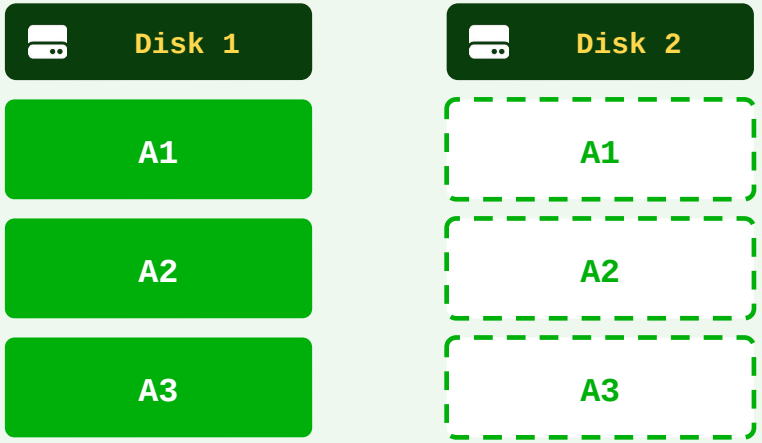
either copy can answer a read

**- 50% usable capacity**

two disks store the data of one

**Use for:** OS drives, logs, critical small datasets.

**RAID 1 — every block written twice**




*Disk 2 is an exact, live copy of Disk 1*

**capacity = 50% · tolerates 1 failure**

**Writes:** slightly slower — both copies must be updated.

RAID 5 — RECOVERY WITHOUT FULL COPIES

# Parity

Parity stores an XOR checksum of the data blocks. If any single disk dies, XOR the survivors and the lost blocks reappear.

```
P  =  A1 XOR A2

A1 =  P  XOR A2      -- rebuild a lost block
```

**+ Cheap redundancy**

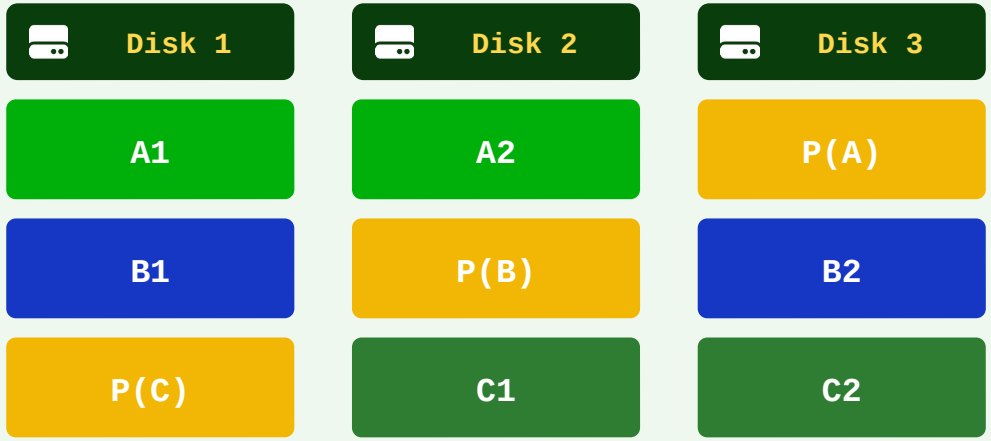
one disk's worth of parity protects all

**- Slower writes**

each write also updates parity

**Distributed:** RAID 5 spreads parity across all disks.

**RAID 5 — parity rotates across disks**




*gold = parity block for that stripe*

**capacity = (n-1)/n · tolerates 1 failure**

**RAID 6:** two independent parities — survives 2 failures.

CHOOSING THE RIGHT LEVEL

# RAID Levels Compared

Each RAID level mixes striping, mirroring and parity differently — pick by how much speed, capacity and safety you need.

| Level | Technique | Min disks | Survives | Usable space | Best for |
|---|---|---|---|---|---|
| RAID 0 | Striping only | 2 | 0 failures | 100% | speed & scratch data |
| RAID 1 | Mirroring only | 2 | 1 failure | 50% | OS & critical data |
| RAID 5 | Striping + parity | 3 | 1 failure | (n-1)/n | general-purpose servers |
| RAID 6 | Striping + dual parity | 4 | 2 failures | (n-2)/n | large storage arrays |
| RAID 10 | Mirrors, then striped | 4 | 1 per pair | 50% | databases & OLTP |

**Rule of thumb:** RAID 10 for databases (fast, safe writes) · RAID 5/6 for bulk storage · RAID 0 only for disposable data.

KEY TAKEAWAYS

# Recap

### Storage Hierarchy

Cache → RAM → SSD → HDD → tape

### Storage Structures

Files → blocks (4–16 KB) → records

### Striping

Parallel blocks across disks — speed

### Mirroring

Full duplicates — reliability

### Parity

XOR checksums — cheap recovery

### RAID Levels

0 speed · 1 mirror · 5/6 parity · 10 both

# Thank You

**Md. Rakib Trofder**
Lecturer, Bangladesh Army University of Science & Technology